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Resistance profiles and genotyping of extended-spectrum beta-lactamase (ESBL) -producing and non-ESBL-producing *E. coli* and *Klebsiella* from retail market fishes

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ABSTRACT

Studies on antimicrobial resistance (AMR) profiles and epidemiological affirmation for AMR transmission are limited in fisheries and aquaculture settings. Since 2015, based on Global Action Plan on AMR by World Health Organization (WHO) and World Organization for Animal Health (OIE), several initiatives have been under taken to enhance the knowledge, skills and capacity to establish AMR trends through surveillance and strengthening of epidemiological evidence. The focus of this study was to determine the prevalence of antimicrobial resistance (AMR), its resistance profiles and molecular characterization with respect to phylogroups, antimicrobial resistance genes (ARGs), virulence genes (VGs), quaternary ammonium compounds resistance (QAC) genes and plasmid typing in retail market fishes. Pulse field gel electrophoresis (PFGE) to understand the genetic lineage of the two most important Enterobacteriaceae members, *E. coli* and *Klebsiella* sp. was performed. 94 fish samples were collected from three different sites viz., Silagrant (S1), Garchuk (S2) and North Guwahati Town Committee (NGTC) Region (S3) in Guwahati, Assam. Out of the 113 microbial isolates from the fish samples, 45 (39.82%) were *E. coli*; 23 (20.35%) belonged to *Klebsiella* genus. Among *E. coli*, 48.88% ($n = 22$) of the isolates were alerted by the BD Phoenix M50 instrument as ESBL, 15.55% ($n = 7$) as PCP and 35.55% ($n = 16$) as non-ESBL. *E. coli* (39.82%) was the most prevalent pathogen among the Enterobacteriaceae members screened and showed resistance to ampicillin (69%) followed by cefazoline (64%), cefotaxime (49%) and piperacillin (49%). In the present study, 66.66% of *E. coli* and 30.43% of *Klebsiella* sp. were categorized as multi drug resistance (MDR) bacteria. CTX-M-gp-1, with CTX-M-15 variant (47%), was the most widely circulating beta-lactamase gene, while other ESBL genes bla_{TEM} (7%), bla_{SHV} (2%) and $bla_{OXA\text{-}1\text{-}like}$ (2%) were also identified in *E. coli*. Out of the 23 isolates of *Klebsiella*, 14(60.86%) were ampicillin (AM)-resistant (11(47.82%) *K. oxytoca*, 3(13.04%) *K. aerogenes*), whereas 8(34.78%) isolates of *K. oxytoca* showed intermediate resistance to AM. All *Klebsiella* isolates were susceptible to AN, SCP, MEM and TZP, although two *K. aerogenes* were resistant to imipenem. DHA and LAT genes were detected, respectively, in 7(16%) and 1(2%) of the *E. coli* strains while a single *K. oxytoca* (4.34%) isolate carried MOX, DHA and $bla_{CMY\text{-}2}$ genes. The fluoroquinolone resistance genes identified in *E. coli* included *qnrB* (71%), *qnrS* (84%), *oqxB* (73%) and *aac(6)-Ib-cr* (27%); however, in Klebsiella, these genes, respectively, had a prevalence of 87%, 26%, 74% and 9%. The *E. coli* isolates belonged to phylogroup A(47%), B1 (33%) and D(14%). All of the 22(100%) ESBL *E. coli* had chromosome-mediated disinfectant resistance genes viz., *ydgE, ydgF, sugE(c), mdfA* while 82% of ESBL *E. coli* had *emrE*. Among the non-ESBL *E. coli* isolates, 87% of them showed the presence of *ydgE, ydgF* and *sugE(c)* genes, while 78% of the isolates had *mdfA* and 39% had *emrE* genes respectively. 59% of the ESBL and 26% of the non-ESBL *E. coli* had showed the presence of *qacEΔ1*. The *sugE(p)* was present in 27% of the ESBL-producing *E. coli* and in 9% of non-ESBL isolates. Out of the 3 ESBL-producing *Klebsiella* isolates, 2(66.66%) *K. oxytoca* isolates were found harboring plasmid-mediated *qacEΔ1* gene while one (33.33%) *K. oxytoca* isolate had *sugE(p)* gene. IncFI was the most prevalent plasmid type detected in the isolates studied, with A/C (18%), P (14%), X, Y (9% each) and I1-Iγ (14%, 4%). 50% ($n = 11$) of the ESBL

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and 17% ($n = 4$) of the non-ESBL *E. coli* isolates harboured IncFIB and 45% ($n = 10$) ESBL and one (4.34%) non-ESBL *E. coli* isolates harboured IncFIA. Dominance of *E. coli* over other Enterobacterales and diverse phylogenetic profiles of *E. coli* and *Klebsiella* sp. suggests the possibility of contamination and this may be due to compromised hygienic practices along the supply chain and contamination of aquatic ecosystem. Continuous surveillance in domestic markets must be a priority in addressing antimicrobial resistance in fishery settings and to identify any unwarranted epidemic clones of *E. coli* and Klebsiella that can challenge public health sector.

1. Introduction

The microbial communities in different environments and their resistomes have been the subject of intense study in recent times. The emergence of AMR is a natural phenomenon; however, the misuse and overuse of antibiotics, paucity of clean water, sanitation and compromised hygienic practices can act as main drivers for the development of antimicrobial-resistant pathogens (World Health Organization, 2021). The widespread use of antibiotics as prophylactics in settings such as livestock and aquaculture reiterate the importance of the 'One Health' approach in addressing the issue of antimicrobial resistance (AMR). Over the years, multi-drug and pan-drug resistant strains of many pathogenic bacteria have emerged (Gashaw et al., 2018; Ozma et al., 2022). Among Gram-negative bacteria (GNB) alone, a number of resistance genes are in circulation, causing these bugs to have impounding consequences on therapy and environment. Extended-spectrum beta-lactamase (ESBL) genes such as $bla_{\text{CTXM-1}}$, bla_{TEM}, $bla_{\text{CTXM-8}}$ and carbapenemase genes viz., bla_{NDM}, bla_{IMP}, bla_{OXA}, bla_{VIM} are some of the prominent resistance genes that are circulating in Gram-negative bacilli members: *K. pneumoniae, A. baumannii, P. aeruginosa, E. coli* and *Enterobacter* sp. (Elbadawi et al., 2021; Manandhar et al., 2020). ESBLs are enzymes that confer resistance to first-, second- and third- generation cephalosporins and the monobactams, besides being susceptible to carbapenems, cephamycins and other beta-lactamase inhibitors (Ramadan et al., 2019). Transmission of ESBL-producing bacteria from animals to humans can occur through horizontal gene transfer, direct contact, food or exposure to environmental runoff (Hoelzer et al., 2017). However, spread of food-borne pathogens is primarily due to consumption of contaminated food/poor hygienic conditions/handling practices. According to WHO 2015 report, unsafe food and unhygienic conditions are responsible for around 550 million people to fall ill and causing 230,000 deaths every year (World Health Organization, 2015). Enterobacteriaceae members especially *Escherichia coli* and *Klebsiella* species are known to cause various infections in animals and humans alike. *E. coli* that has mammalian intestine as its natural habitat is capable of causing intra- and extra-intestinal infections. It has evolved resistance to major antibiotic classes and it is projected that, by 2050, this pathogen would emerge as a superbug, posing serious threat to public health (O'Neill, 2014). A recent analysis on the association of AMR in global deaths in 2019 has implicated third-generation cephalosporin resistant *E. coli* and carbapenem resistant *K. pneumoniae* among the leading bacterial pathogens (Murray et al., 2022).

India has gradually increased its share of marine products export to almost 121 countries with a cumulative annual growth rate of 8.23% in the past decade. Based on the seafood export data released by the Marine Products Exports Development Authority (MPEDA, n.d., https://mpeda.gov.in/), India has exported over 12 lakh tonnes of seafood an all-time high of $7.74 billion in the financial year 2021–2022. Prevalence of antibiotic-resistant bacteria and antibiotic-resistant genes has been very crucial food safety concerns in recent times (Sheng and Wang, 2021). In this context, a study on pervasiveness of AMR in fish and fishery environment holds utmost significance.

The objective of the study was to determine the antimicrobial resistance and molecular epidemiologies of *E. coli* and *Klebsiella* sp. (*Klebsiella oxytoca*, *Klebsiella aerogenes*, *Klebsiella pneumoniae*) occurring in fish. For this, a collection of isolates was subjected to antimicrobial susceptibility tests (AST), screening for antibiotic and disinfectant resistance genes, virulence profiling, phylogrouping, PCR-based replicon typing (PBRT) and molecular sub-typing by pulsed- field gel electrophoresis (PFGE). A comparative analysis of the above characteristics in ESBL-producing and non-ESBL- producing isolates was also evaluated.

2. Materials and methods

2.1. Study sites and sample collection

This study was part of an investigation to screen antibiotic resistance among Enterobacteriaceae in fishery environment, identify AMR drivers and investigate the transmission dynamics between animals and humans in three specific locations viz., Silagrant (S1), Garchuk (S2) and North Guwahati Town Committee (NGTC) Region (S3) in Guwahati, the capital city of Assam state, India. Different types of edible fishes that were sold in retail markets and by street fish vendors were collected/purchased in the month of February 2020 Only one location was chosen for the collection on any particular day and fish samples were transferred in a sterile polythene bag (HiMedia, India) and were maintained in cold conditions. Preliminary information from vendors suggested that these fishes had either been captured from rivers or beels (ponds/lakes) in the locality or imported from nearby states. Baseline details such as the source of fish, its taxonomical identity, and geographical co-ordinates of the collection site were documented. A total of 94 fish samples were collected from the specified locations and were transported on ice to the laboratory for initial processing. Pre-processing of the samples was carried out at the Indian Council for Agricultural Research (ICAR)-National Research Centre for Pig (NRCP), Rani, Guwahati. Pre-processing included the initial preparation (maceration) of the fish sample and inoculation onto the enrichment media. Identification and molecular investigation of the isolates were performed at the Microbiology, Fermentation and Biotechnology Division of ICAR-Central Institute of Fisheries and Technology (CIFT), Cochin, Kerala.

2.2. Bacterial isolation, identification and antimicrobial susceptibility testing (AST)

Surface of the fishes were wiped clean with 70% alcohol and the gut sample along with muscle tissue was pooled from each fish sample and macerated aseptically. Macerated samples were inoculated onto Enterobacteriaceae Enrichment Broth (EE Broth Mossel) (BD Difco, USA) and were incubated at 37 °C for 24 h. MacConkey agar (BD Difco, USA) containing 1 μg/mL cefotaxime (Sigma Aldrich, USA) was used for selecting possible ESBL-producing isolates. Similarly, susceptible isolates were selected from MacConkey agar plates without cefotaxime. Typical pink dry colony indicative of lactose-fermenting characteristics of *E. coli* and pink mucoid colony indicative of *Klebsiella* sp. were selected and sub-cultured onto Eosin-Methylene Blue agar (EMB, pH 7.2 ± 0.2) (BD Difco, USA). Colonies with green metallic sheen and pink to purple-coloured colonies from EMB plates were harvested. Other Enterobacteriaceae colonies which grew on both MacConkey and EMB agar plates were also isolated. A single colony for each distinct morphological trait was selected from each sample. Pure colonies were sub-cultured on BD Tryptic Soy Agar (TSA, pH 7.3 ± 0.2) and stored in Tryptic Soy Broth (TSB) (BD Difco, USA) containing 20% glycerol at −80 °C for subsequent analyses.

Identification and antibiotic susceptibility testing of the pure colonies were performed in specified gram-negative panels (NMIC/ID-55) in BD Phoenix™ M50 automated system (BD Diagnostics, USA), following the manufacturer's protocol. The panels had twenty-one major antibiotics representing thirteen antimicrobial classes along with ESBL detection. Based on the bacterial growth response to second or third generation cephalosporins, in the presence or absence of the clavulanic acid, a beta-lactamase inhibitor, the strain was interpreted by BD Phoenix M50 system as an ESBL producer. Also, isolates with minimum inhibitory concentrations (MIC) suggestive of intermediate or resistant phenotype for carbapenems, were alerted as 'potential carbapenemase producer' (PCP). The strains which were sensitive to 3rd generation cephalosporins were referred to as non-ESBL in the study. The antibiotics that were tested included amikacin (AN, 8-32 μg/mL), amoxicillin/clavulanate (AMC, 4/2–16/8 μg/mL), ampicillin (AM, 4–16 μg/mL), aztreonam (ATM, 2–16 μg/mL), cefazolin (CZ, 4–16 μg/mL), cefepime (CPM, 2–16 μg/mL), cefoperazone/sulbactam (SCP, 0.5/8–16/8 μg/mL), cefotaxime (CTX, 4–32 μg/mL), cefoxitin (FOX, 4–16 μg/mL), ceftazidime (CAZ, 1–16 μg/mL), chloramphenicol (C, 4–16 μg/mL), ciprofloxacin (CIP, 0.5–2 μg/mL), gentamicin (GN, 2–8 μg/mL), imipenem (IPM, 1–8 μg/mL), levofloxacin (LVX, 1–4 μg/mL), meropenem (MEM, 1–8 μg/mL), piperacillin (PIP, 4–64 μg/mL), piperacillin/tazobactam (TZP, 4/4–64/4 μg/mL), tetracycline (TE, 2–8 μg/mL), trimethoprim/sulfamethoxazole (SXT, 0.5/9.5–2/38 μg/mL). MIC of the isolates against each antibiotic was interpreted strictly adhering to Clinical and Laboratory Standards Institute (CLSI) guidelines (CLSI, 2020). *E. coli* (ATCC 25922) was used as reference strain. Molecular identification of the two important pathogens *E. coli* and *Klebsiella* sp. was performed according to the conditions mentioned by Thong and Yu, 2009; Bhargava, 2019; Chander et al., 2011. The PCP isolates were tested for confirming carbapenem-resistant Enterobacteriaceae (CRE) following Dallenne et al., 2010 and Kaase et al., 2012. The details of the primers used and the reaction conditions of all the polymerase chain reactions performed in this study are provided in Supl. Table 1.

2.3. Phylogrouping of *E. coli*

A quadruplex PCR that determines a specific phylogroup (B1, B2 and F) based on the amplification patterns of *arpA* (400 bp), *chuA* (288 bp), *yjA* (211 bp) and *TspE4.C2* (152bp) and other additional PCR reactions for assigning the isolates to either A, C, D, E and clade I were performed (Clermont et al., 2013).

2.4. Molecular detection of antibiotic resistance genes (ARGs), virulence genes (VGs) and quaternary ammonium compounds (QAC) resistance genes

2.4.1. Detection of ARGs in *E. coli* and *Klebsiella* sp.

Screening for ESBL and plasmid-mediated AmpC beta-lactamase genes were performed according to the multiplex PCR protocols described elsewhere (Dallenne et al., 2010). Multiplex I targeted bla_{TEM}, bla_{SHV} and $bla_{\mathrm{OXA\text{-}1\text{-}like}}$ genes; multiplex II targeted $bla_{\mathrm{CTX\text{-}M}}$-group 1, 2 and 9; and Multiplex III was performed to determine the family-specific plasmid-mediated AmpC beta-lactamases (ACC, FOX, MOX, DHA, CIT and FBC). The strains were further checked for the presence of the most common $bla_{\mathrm{CTX\text{-}M\text{-}15}}$ gene variant (Gundran et al., 2019). Resistance to cephamycin group of drugs is characterized by the plasmid encoding cephamycinase genes like $bla_{\mathrm{CMY\text{-}2}}$ (Bauernfeind et al., 1996; Zhao et al., 2001). Isolates of *E. coli* and *Klebsiella* sp. that showed resistance to cefoxitin and/or had MIC suggestive of 'intermediate/resistant' category for antibiotic-inhibitor combination of amoxicillin-clavulanic acid were tested for the plasmid mediated $bla_{\mathrm{CMY\text{-}2}}$ gene (Kozak et al., 2009).

Based on the phenotypic resistance profiles, isolates were screened for genes conferring resistance to quinolones (*qepA, oqxA, oqxB, qnrA, qnrB, qnrS* and *aac(6)-Ib-cr*), chloramphenicols (*cml* and *cat1*), aminoglycosides (*strA*, *strB*, *aadA1, aphA1-1AB* and *aac(3)-IV*), tetracycline (*tetA, tetB* and *tetG*), and sulphonamides (*dfrA1, sul1* and *sul2*). The primer sequences, concentration of each component in the polymerase chain reaction (PCR) mixture and amplification conditions are given in Supl. Table 1.

2.4.2. PCR screening of virulence genes in *E. coli* and *Klebsiella* sp.

A single multiplex PCR was carried out to identify diverse loci associated with enterotoxigenic (ETEC), enteropathogenic (EPEC), enteroinvasive (EIEC) and shiga toxin (STEC) producing *E. coli*. Screening of virulence genes in *E. coli* and *Klebsiella* sp. were performed. The primers used in the study and the amplification conditions employed were according to López-Saucedo et al. (2003). The cycling conditions included a two-step denaturation process with temperatures of 50 °C for 2 min and 95 °C for 5 min (1×). This was followed by 40 cycles of 95 °C for 45 s, 50 °C for 45 s and 72 °C for 45 s. The PCR was completed with a final extension at 72 °C for 10 min. The expected amplicons for the corresponding genes are as follows: *LT* (450 bp), *ST* (190 bp), *bfpA* (324 bp), *eaeA* (384 bp), *stx1* (150 bp), *stx2* (255 bp) and *ial* (650 bp).

A multiplex PCR for the determination of capsular serotypes and virulence genes in *Klebsiella* sp. was performed. The serotype specific genes *magA* (K1) and K2, and the virulence specific genes *ybt, mrkD, entB, rmpA, kfu, allS* and *iutA* were screened by using primers and amplification conditions as described by Compain et al., 2014. Though the primers mentioned were exclusively designed for *K. pneumoniae*, we have tested in *K. oxytoca*, *K. aerogenes* and *K. pneumoniae*. The reaction conditions included an initial denaturation at 95 °C for 15 min, 30 cycles of denaturation at 94 °C for 30s, annealing at 60 °C for 90s, extension at 72 °C for 60s, and a final extension at 72 °C for 10 min.

2.4.3. PCR amplification of disinfectant resistance genes

All the isolates were screened for various quaternary ammonium compounds (QAC) resistance genes (Wu et al., 2015). This included the plasmid and/or integron encoded resistance genes (*qacE, qacEΔ1, qacF, qacG* and *sugE(p)*) and the chromosome-encoded genes (*sugE(c), emrE, ydgE/ydgF* and *mdfA*).

All multiplex and uniplex PCRs were performed in a final volume of 25 μL that comprised 1× Jumpstart RedTaq ReadyMix (Sigma-Aldrich, USA), 2 μL of DNA extract and varied concentrations of primers. PCR products were analyzed by electrophoresis on 2% agarose gel containing ethidium bromide (1 μg/mL) in 1× TAE buffer (40 mM Tris–HCl, 1.18 mL acetic acid, 2 mM EDTA, pH 8.0). The image was documented using BioRad Automated Gel Imaging System (USA). All the oligonucleotides required for the study were synthesized by Merck, India and the details of the primer sequences used in the study is given in Supl. Table 1.

2.5. PCR-based replicon typing (PBRT)

To identify the replicons of incompatibility plasmid groups among *E. coli* and *Klebsiella sp,* a PCR-based replicon typing (inc/rep) which involved 5 multiplex and 3 uniplex reactions were performed (Carattoli et al., 2005). For this purpose, a series of primers (18 pairs) recognizing the FIA, FIB, FIC, HI1, HI2, I1-Iγ, L/M, N, P, W, T, A/C, K, B/O, X, Y, F, and FIIA replicons were used. All five multiplex PCR were performed with the following conditions: one cycle of initial denaturation at 94 °C for 5 min followed by 30 cycles of denaturation at 94 °C for 1 min, annealing at 60 °C for 30s and elongation at 72 °C for 1 min. Final extension at 72 °C for 5 min was set. Same amplification scheme was adopted for F-simplex except for the annealing temperature of 52 °C.

2.6. Pulsed-field gel electrophoresis (PFGE)

Pulsed-Field Gel Electrophoresis was carried out following the standard operating procedure by PulseNet International. Agarose plugs containing genomic DNA of *E. coli* or *Klebsiella* sp. were digested with *XbaI* (50 U/sample, New England BioLabs) at 37 °C for 1.5–2 h. Plugs were loaded onto 1% Megabase agarose gel (Bio-Rad, United States) and the run was carried out in 0.5× TBE (Tris Boric acid-EDTA) buffer at 14 °C in a CHEF-Mapper XA device (Bio-Rad, United States). Electrophoretic conditions included a constant voltage of 6 V/cm, a run time of 19 h with pulse time ranging from 6.76 to 35.38 s and an angle of 120°. The Braenderup H9812 Salmonella serotype was used as the standard molecular marker. The gels were stained in ethidium bromide (1 μg/mL) and documented. Images were imported to Bionumerics software ver. 7.6 (AppliedMaths, Belgium) and analyzed. Cluster analysis was performed using the Unweighted Pair Group Method with Arithmetic mean (UPGMA) and Dice coefficient with 1% tolerance and optimization. Pulsotypes and cluster assignment were resolved by setting the cutoff value to ≥85%. If the strains possessed 100% similarity, they were considered a single strain while clonal relatedness was assumed if the PFGE profiles showed ≥85% as previously described (Tenover et al., 1995).

3. Results

3.1. Sample details and ethnographic data

A total of 94 fish samples were used in the present study. Of these, 28 were collected from site-1 (Silagrant), 33 each from site 2 (Garchuk) and site 3 (NGTC). The details of the fish species, source and the geographical co-ordinates of each collection site have been mentioned (Supl. Table 2). Fishes sold in the retail markets were mostly farmed fishes, or had been imported from neighboring states. Fishes sold in local streets were either beel- or river-caught that included indigenous species.

3.2. Bacterial identification, ESBL detection and AST

A total of 113 microbial isolates belonging to Enterobacteriaceae have been recovered from the fish samples screened. Identified species included *E. coli* (39.82%), *Klebsiella oxytoca* 16.81%), *K. aerogenes* (2.65%), *K. pneumoniae* (0.88%), *Citrobacter freundii* (19.47%), *Citrobacter farmeri* (1.77%), *Citrobacter braakii* (7.08%), *Citrobacter werkmanii* (1.77%), *Enterobacter cloacae* (3.54%), *Hafnia alvei* (2.65%), *Pantoea agglomerans* (0.88%), *Moellerella wisconsensis* (0.88%), *Kluyvera cryocrescens* (0.88%) and *Proteus vulgaris* (0.88%). *E. coli* confirmation at molecular level produced the target amplicons of 603 bp for *uidA* gene and 903 bp of *phoA* gene. Similarly, all *Klebsiella* isolates were confirmed by the PCR amplification of *gyrA* (441 bp) gene, a locus conserved in *Klebsiella* genus. The *pehX* (343 bp) gene amplification confirmed the isolates as *K. oxytoca* and *rpoB* (108 bp) gene as *K. pneumoniae*.

The distribution of all bacterial species isolated from three studied sites is represented in Table 1. However, considering the relevance and pathogenicity, we have considered only *E. coli* and *Klebsiella sp* for further studies. Out of the 113 isolates, 45 (39.82%) were *E. coli* and 23 (20.35%) belonged to *Klebsiella* genus. Among *E. coli*, 48.88% (*n* = *22*) of the isolates were alerted by the BD Phoenix M50 instrument as ESBL, 15.55% (*n* = *7*) as PCP and 35.55% (*n* = *16*) as non-ESBL. In *Klebsiella* sp. we found three ESBL-producing *K. oxytoca* isolates corresponding to 13.04% (*n* = *3*), while 21.73% (*n* = *5*) of *K. oxytoca* and 13.04% (*n* = *3*) of *K. aerogenes* were alerted as probable carbapenemase producers based on MICs of imipenem and meropenem. On a positive note, 47.82% (*n* = *11)* of *K. oxytoca* and 4.34% (*n* = *1*) of *K. pneumoniae* isolates from this study were non-ESBL producers.

Among the *E. coli* strains, 31(69%) showed resistance to ampicillin, cefazoline (29, 64%), cefotaxime (22, 49%) and piperacillin (22, 49%). Similarly, 19(42%) isolates showed resistance to aztreonam and 17 (38%) to cefepime and tetracycline. Ciprofloxacin resistance was seen in 13(29%) and trimethoprim/sulfamethaxazole in 12(27%) isolates. All *E. coli* isolates were susceptible to amikacin and 96–98% of the isolates were susceptible to SCP, MEM, TZP, C and GM. Notably, one *E. coli* isolate from site 2 (Garchuk) was resistant to 16/21 antibiotics tested, with a multiple-antibiotic resistance (MAR) index of 0.76. Overall, 30 (66.67%) of the *E. coli* isolates showed resistance to antibiotics belonging to 3 or more antimicrobial drug classes thus classifying them as multi-drug resistant (MDR) isolates. With regard to antibiotic resistance, 14(61%) of the *Klebsiella*, isolates showed resistance to ampicillin and cefazolin. Out of the 23 isolates of *Klebsiella*, 14 were AM-resistant (11 *K. oxytoca*, 3 *K. aerogenes*), whereas 8 isolates of *K. oxytoca* showed intermediate resistance to AM. Surprisingly, one *K. pneumoniae* isolate was susceptible to all the tested antibiotics including ampicillin and piperacillin. Among the other antibiotics, amoxicillin/clavulanate resistance was observed in 6(26%) while cefoxitin and tetracycline resistance was detected in 5(22%) isolates. Three *K. oxytoca* strains showed cefotaxime resistance while the rest of them showed intermediate resistance. Chloramphenicol resistance was detected in 3 *K. oxytoca* while aztreonam and trimethoprim/sulfamethaxazole resistance was observed in 2 isolates. All *Klebsiella* isolates were susceptible to AN, SCP, MEM and TZP, although two *K. aerogenes* were resistant to imipenem. Intermediate MICs were recorded for levofloxacin and ciprofloxacin in all but two isolates of *K. aerogenes*. Gentamicin resistance was seen only in one *K. oxytoca*. Of the 23 isolates of *Klebsiella sp,* 7 (30.43%) were classified as MDR as they showed resistance to antibiotics belonging to ≥3 antimicrobial drug classes. (Intrinsic ampicillin resistance of *Klebsiella* sp. has not been considered to identify MDR). Table 2 shows the total number of susceptible and resistant isolates to each antibiotic tested in the present study. MIC value of all the isolates against 21 antibiotics tested and MDR status have been represented in Supl. Table 3.

Comparison of antibiotic resistance profile in three selected sites is shown in Fig. 1. High prevalence of cephalosporin resistance was observed in *E. coli* and *Klebsiella* from all three sites, with *E. coli* having higher antibiotic resistance pattern compared to *Klebsiella* species. Both *E. coli* and *Klebsiella* sp. isolated from site 1 (Silagrant) have shown resistance to majority of the antibiotics and multi-drug resistance was prominent in *E. coli* isolates from site 2 and 3.

3.3. Phylogroup assignment of E. coli

In the present study, ESBL-producing *E. coli* isolates belonged to five different phylogroups, phylogroup A (45%) being the predominant one. Isolates were equally distributed to Group B1 and D with a prevalence rate of 14%. One ESBL-producing isolate each from site-1 and site-2 belonged to phylogroup C and E respectively while two ESBL-producing isolates from site-3 belonged to phylogroup E. However, the non-ESBL isolates belonged to only two phylogroups namely A (48%) and B1(52%). On comparison, we found that site 1 and 2 had representations in phylogroups A, B1, C, D and E, while site 3 had group A, B1, D and F. Overall, the majority of *E. coli* isolates from this study belonged to group A(47%) and phylogroup B1(33%).

Table 1
Distribution of different species of bacteria in three study sites.

Bacteria	No. of isolates (*n*)			Total (%)
	Site 1 (Silagrant)	Site 2 (Garchuk)	Site 3 (NGTC)	
E. coli	9	24	12	39.82
K. oxytoca	12	1	6	16.81
K. aerogenes	2	0	1	2.65
K. pneumoniae	0	0	1	0.88
C. freundii	7	8	7	19.47
C. farmeri	1	1	0	1.77
C. braakii	5	0	3	7.08
C. werkmanii	1	1	0	1.77
E. cloacae	2	0	2	3.54
H. alvei	3	0	0	2.65
P. agglomerans	0	1	0	0.88
M. wisconsensis	0	0	1	0.88
K. cryocrescens	1	0	0	0.88
P. vulgaris	0	1	0	0.88

Table 2
Antibiotic susceptibility [SIR*- No. of isolates (%)]- *E. coli* and *Klebsiella* sp. *SIR-Susceptibility, Intermediate, Resistant based on CLSI, 2020.

Antibiotics	Abbreviation	Range (MIC, µg)	*E. coli* Susceptibility			*Klebsiella* sp. Susceptibility		
			Resistant No. (%)	Intermediate No. (%)	Sensitive No. (%)	Resistant No. (%)	Intermediate No. (%)	Sensitive No. (%)
β- lactams								
Amoxicillin/Clavunate	AMC	4/2–16/8	9(20)	4(9)	32(71)	6(26)	2(9)	15(65)
Ampicillin	AM	4–16	31(69)	14(31)	0	14(61)	8(35)	1(4)
Aztreonam	ATM	2–16	19(42)	2(4)	24(53)	2(9)	0	21(91)
Cefazolin	CZ	4–16	29(64)	16(36)	0	14(61)	9(39)	0
Cefepime	FEP	2–16	17(38)	3(7)	25(56)	2(9)	1(4)	20(87)
Cefoperazone/Sulbactam	SCP	0.5/8–16/8	0	1(2)	44(98)	0	1(4)	22(96)
Cefotaxime	CTX	4–32	22(49)	0	23(51)	3(13)	20(87)	0
Cefoxitin	FOX	4–16	9(20)	0	36(80)	5(22)	0	18(78)
Ceftazidime	CAZ	1–16	11(24)	6(13)	28(62)	1(4)	0	22(96)
Imipenem	IPM	1–8	1(2)	7(16)	37(82)	2(9)	5(22)	16(70)
Meropenem	MEM	1–8	1(2)	0	44(98)	0	1(4)	22(96)
Piperacillin	PIP	4–64	22(49)	4(9)	19(42)	3	0	20
Piperacillin/Tazobactam	TZP	4/4–64/4	1(2)	0	44(98)	0	0	23(100)
Quinolones and Fluoroquinolones								
Ciprofloxacin	CIP	0.5–2	13(29)	32(71)	0	2(9)	21(91)	0
Levofloxacin	LVX	1–4	10(22)	33(73)	0	0	23(100)	0
Aminoglycosides								
Amikacin	AN	8–32	0	0	45(100)	0	0	23(100)
Chloramphenicol	C	4–16	2(4)	0	43(96)	3(13)	0	20(87)
Gentamicin	GM	2–8	1(2)	0	44(98)	1(4)	0	22(96)
Tetracyclines								
Tetracycline	TE	2–8	17(38)	0	28(62)	5(22)	0	18(78)
Sulfonamides								
Trimethoprim/ Sulfamethoxazole	SXT	0.5/9.5–2/38	12(27)	0	33(73)	2(9)	0	21(91)

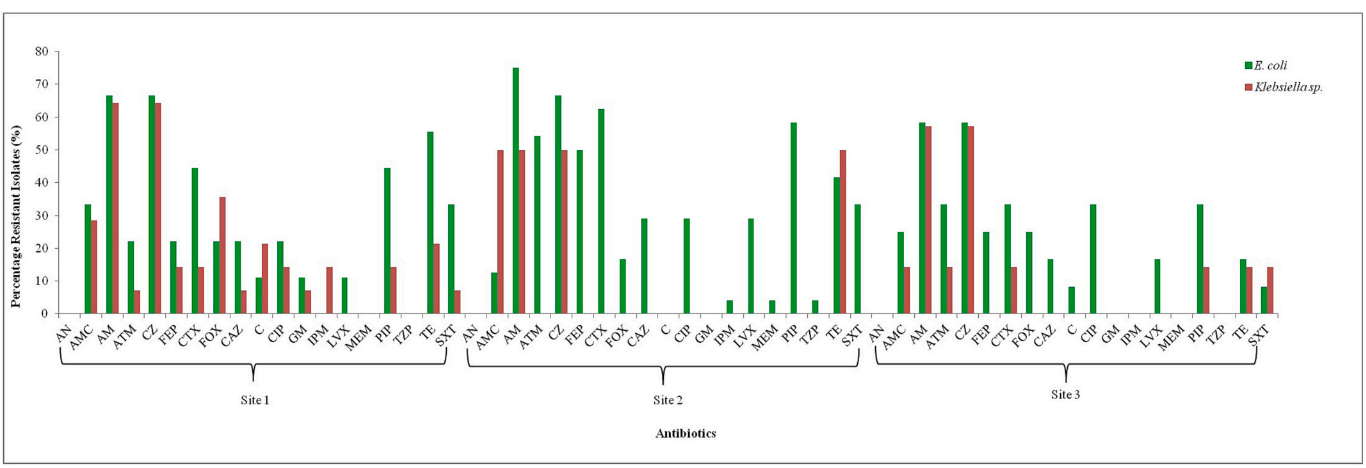

Fig. 1. Resistance pattern of *E. coli* and *Klebsiella* sp. isolates from three different study sites.

3.4. Molecular characterization of resistance genes

Screening for beta-lactamase genes were performed for those isolates that were alerted as ESBL-producers. We found that CTX-M-gp-1 was the predominant ESBL gene in *E. coli* (47%), while ESBL-producing *K. oxytoca* carried CTX-M-gp-1 and CTX-M-gp-9. All ESBL-producing *E. coli* isolates carried *bla*$_{\text{CTX-M-15}}$ gene except for one isolate. Two ESBL-producing *E. coli* isolates that carried *bla*$_{\text{CTX-M-15}}$ gene did not show amplification with CTX-M-gp-1 primer. Similarly, among three ESBL-producing *K. oxytoca*, one isolate was positive for CTX-M-15 gene and belonged to CTX-M- group-1. The second isolate had tested positive with CTX-M-9 while the third showed amplification with CTXM-14 indicating that the latter two belonged to CTX-M- group 9. ESBL-*E. coli* isolates when tested for CTX-M-gp-1 in general and CTX-M-15 gene variant in specific, 19 out of 22 harboured both CTX-M-gp-1 and CTX-M-15 genes while two isolates had only CTX-M-15 gene variant and the other one was positive for CTX-M-gp-1. When we checked the antimicrobial susceptibility test values of these three isolates, the isolate that showed amplification for CTX-M-gp-1 was susceptible to ceftazidime; one isolate with CTX-M-15 gene variant was observed with intermediate MIC value for cefotaxime while all three were having resistant MIC values for aztreonam. Not all ESBL *E. coli* isolates that showed the presence of CTX-M-15 and CTX-M-gp-1 cefotaximases conferred phenotypic resistance to ceftazidime and aztreonam. One ESBL-*E. coli* was sensitive to aztreonam despite being positive for CTX-M-gp-1 and CTX-M-15. In ESBL-*K. oxytoca* isolates, presence of either one of the genes representing CTX-M-gp-9 or CTX-M-14, was associated with the phenotypic susceptibility/resistance to aztreonam.

Other ESBL genes that were identified in *E. coli* were bla_{TEM} (3, 7%), bla_{SHV} (1, 2%) and $bla_{\text{OXA-1-like}}$ (1, 2%); in *Klebsiella*, only one isolate was positive for bla_{TEM} and no other isolates carried bla_{SHV} or $bla_{\text{OXA-1-like}}$ resistance genes. Among the plasmid-mediated AmpC beta-lactamase genes, both *E. coli* and *Klebsiella* had few representative genes: 7 (16%) *E. coli* isolates carried DHA and one was identified with LAT gene. Among *Klebsiella*, one *K. oxytoca* which was alerted as PCP showed MOX, DHA and $bla_{\text{CMY-2}}$ genes. The fluoroquinolone resistance genes identified in *E. coli* included *qnrB* (32, 71%), *qnrS* (38, 84%), *oqxB* (33, 73%) and *aac(6)-Ib-cr* (12, 27%); however, in *Klebsiella*, these genes, respectively, had a prevalence of (20 isolates, 87%), (6)26%, (17)74% and (2)9%. The chloramphenicol resistance gene *cmlA* was detected only in one isolate of *E. coli* and *Klebsiella*. Among the tetracycline resistance genes, only *tetA* was present in both *E. coli* and *Klebsiella* with an incidence of 33% and 4%, respectively. Only two aminoglycoside resistance genes were identified; *strB* gene detected in one isolate of *Klebsiella* and *aphA1-1AB* detected in each of *E. coli* and *Klebsiella*. Three resistance genes such as *dfrA1* (1, 4%), *sul1* (2, 13%), and *sul2* (1, 4%) conferring resistance to SXT were identified in *Klebsiella*, while (12, 27%) of the *E. coli* isolates were *sul1*-positive and (8)18% were *sul2*-positive. The distribution of AMR genes in *E. coli* and *Klebsiella* has been represented in Fig. 2.

Seven *E. coli* and eight *Klebsiella* isolates were tested for carbapenemase-encoding genes based on the phenotypic results. The NMIC/ID panel used for identification and antimicrobial susceptibility test alerted the isolates of *E. coli* ($n = 7$) and *Klebsiella* ($n = 8$) as potential carbapenamase producers. Despite the fact that these isolates exhibited either intermediate- or complete-resistance to imipenem and meropenem, none of the PCP strains carried any carbapenemase resistance genes (IMP, VIM, NDM, GES, VEB, PER, KPC) and were confirmed non-CRE.

Multiplex PCR that was employed for the detection of virulence genes specific to the major pathotypes of *E. coli* (ETEC, EPEC, EIEC and STEC) yielded amplification of two loci. Among the loci tested, one *E. coli* isolate, WP5–29 was positive for *LT*(450 bp) confirming the strain's identity as ETEC type, and another isolate (WP5–28) was positive for *eaeA*(384 bp). The *eaeA*-positive strain did not show any co-existence of other virulence genes. In *Klebsiella* sp. none of the strains harboured the capsular serotypes K1 or K2. Among the other virulence genes, *ybtS* gene was detected in 56.5% ($n = 13$) of the *K. oxytoca* isolates while *rmpA* gene was found in 60.8% of the *Klebsiella* isolates that included 12 *K. oxytoca* and 2 *K. aerogenes* isolates.

In the present study, the prevalence of disinfectant resistance genes (both chromosome- and plasmid-mediated) was higher in *E. coli* than in *Klebsiella* as seen in Fig. 3. When compared to plasmid-encoded QAC resistance genes, chromosome-encoded genes were the most widely distributed. All ESBL-producing *E. coli* harboured *ydgE/F*, *sugE(c)* and *mdfA* resistance genes while 82% ($n = 18$) of isolates had *emrE* gene. Among the non-ESBL *E. coli* isolates, 87% ($n = 20$) of them showed the presence of *ydgE*, *ydgF* and *sugE(c)* genes, while 78% (18) of the isolates had *mdfA* and 39% (9) had *emrE* genes. Among the genes associated with mobile-genetic elements, *qacEΔ1* and *sugE(p)* were the two most commonly detected in *E. coli*. Notably, 59% (13) of the ESBL and 26% (6) of the non-ESBL *E. coli* had showed the presence of *qacEΔ1*. The *sugE (p)* was present in 27% (6) of the ESBL-producing *E. coli* and 9% (2) of non-ESBL isolates. *qacE*, *qacF* and *qacG* were completely absent in *E. coli*. The incidence of disinfectant resistance genes in *Klebsiella* was limited. Out of the 3 ESBL-producing *Klebsiella* isolates, 2 *K. oxytoca* isolates were found harboring plasmid-mediated *qacEΔ1* gene while one *K. oxytoca* isolate had *sugE(p)* gene. No other plasmid-encoded genes were present in rest of the *Klebsiella* isolates. Among the chromosome-encoded resistance genes, 10% (2) of both ESBL and non-ESBL isolates carried *sugE(c)* while a limited representation of other genes were noted: *mdfA* (2, 10%), *ydgF*(1, 5%) and *emrE*(1, 5%). Incidentally, the isolates that were detected with *mdfA*, *ydgF* and *emrE* genes were *K. aerogenes* with resistance to amoxicillin/clavulanate, ampicillin and cefazolin in common.

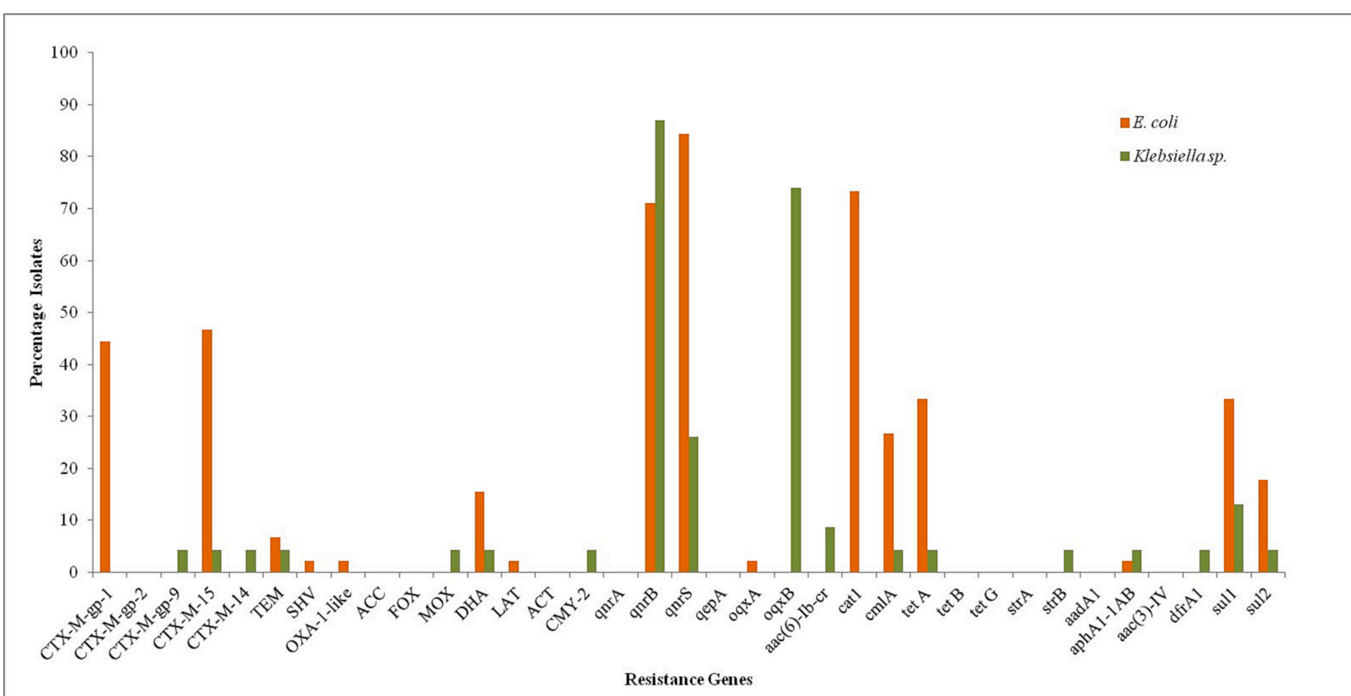

Fig. 2. Distribution of the ARGs present in *E. coli* and *Klebsiella* sp.

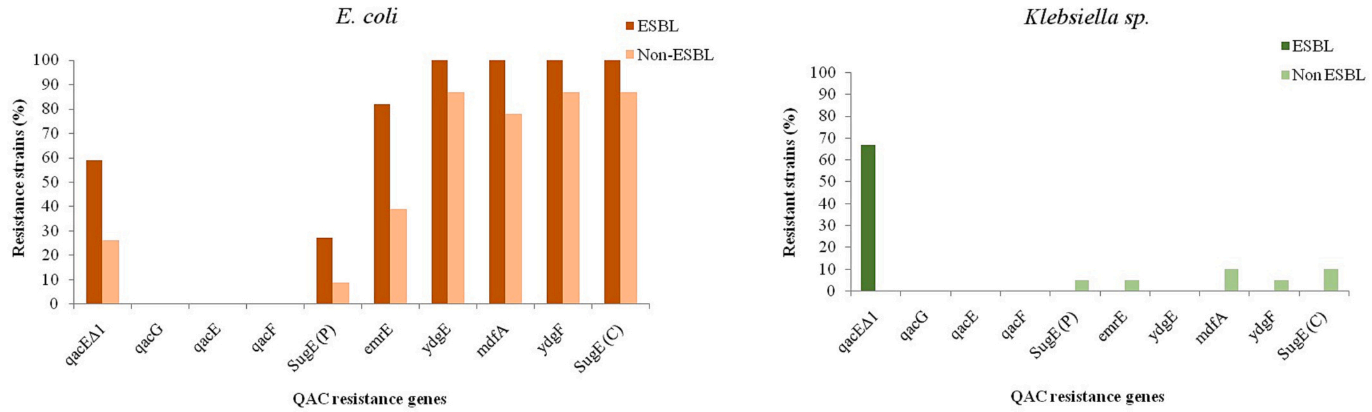

Fig. 3. Disinfectant resistance gene profile of ESBL and non-ESBL producing isolates.

3.5. Distribution of plasmid types

The characterization of Inc./rep plasmid types in *E. coli* strains revealed the predominance of IncFIB-type plasmid in both ESBL and non-ESBL isolates. However, the proportion of IncFIB plasmids detected varied considerably, with 50% ($n = 11$) of the ESBL and 17% ($n = 4$) of the non-ESBL isolates harboring the same. IncFIA was the second dominant plasmid-type found in *E. coli,* with 45% ($n = 10$) ESBL isolates harboring this type. Only one non-ESBL isolate was positive in this category. A/C type representing *repA* genes and P type representing iterons were also detected only in beta-lactamase producing *E. coli,* with a prevalence percentage of 18% and 14%, respectively. Two other plasmid types which shared equal distribution (9%) in ESBL *E. coli* were X and Y plasmid types. These plasmid types i.e., A/C, X and Y type were absent in non-ESBL while only one isolate had P type. I1-Iγ plasmid type was present in both ESBL and non-ESBL isolates (14% and 4% respectively). HI1 type was detected in one non-ESBL isolate. Four ESBL *E. coli* isolates did not carry any of the plasmid types tested. *E. coli* isolates carrying $bla_{\text{CTX-M-15}}$ gene and have shown resistance to quinolones, aminoglycosides and sulfonamides also belonged to IncF plasmid type. However, with this primer set, no plasmid types were detected in the isolates of *Klebsiella*. Table 3 summarizes the distribution of plasmid types in *E. coli* and their corresponding antibiotic resistance gene profile.

3.6. Phylogenetic analysis by PFGE

PFGE analysis of 45 *E. coli* isolates revealed 41 unique genotype profiles. 100% genetic similarity was observed for strains WP4–5 and WP5–6 from site 2 and also between WP5–20 and WP5–21 from site 3. Two ESBL *E. coli* isolates (WP5–22 and WP5–23) from site 3 had a clonal relatedness of 85.7%. One isolate (WP5–16) was not typable by PFGE. The pulsotypes obtained for *Klebsiella* showed 19 distinct profiles. Two sets of isolates had 100% similarity and formed separate clusters. (*K. oxytoca*: WP5–58 and WP5–64 from site 1 and site2, respectively; *K. aerogenes*: WP5–55 and WP5–56 both from site 1). Dendrogram depicting the clonal relatedness of *E. coli* and *Klebsiella* has been shown in Fig. 4.

4. Discussion

ESBL-producing Enterobacteriaceae have been reported from different settings and sources. The majority of studies that have been reported on ESBL from India are associated with clinical settings, veterinary animals and poultry (Abhilash et al., 2010; Lalzampuia et al., 2013; Bhoomika et al., 2016; Mahanti et al., 2018; Tewari et al., 2019). Besides, there is a lack of comparative study of the ESBL and non-ESBL strains with respect to phylogroup, AMR genes, plasmid types and molecular epidemiology existing in a particular environmental niche. Studies pertaining to the frequency and dissemination of antibiotic-resistant bacteria in fish and aquatic environment are also limited (Sivaraman et al., 2021; Khan et al., 2021). Researchers have highlighted the poor hygienic conditions at the landing centres, storage facilities and the open fish markets as the major source of bacterial contamination (Thampuran et al., 2005; Singh et al., 2020). In most cases, quality of the fish gets compromised by the time it reaches the retail markets, aggravating the problem of infestation with pathogens. In this context, this study has focused on the prevalence, antibiotic susceptibility, antimicrobial resistance genes, plasmid genotypes and epidemiology of *E. coli* and *Klebsiella* strains isolated from fishes and also on the genotypic comparison of ESBL and non-ESBL isolates.

The dominance of *E. coli* is quite remarkable (47.9%) over other enterobacterales like *Klebsiella* and *Citrobacter* species in the current study. ESBL-*E. coli* dominates with a prevalence rate of 48.88% while non-ESBL-*Klebsiella oxytoca* isolates accounted for 47.82% of the total *Klebsiella* species identified. We had previously conducted a similar surveillance study in the month of August 2019, and the data showed a higher prevalence rate of ESBL-producing *E. coli* and *Klebsiella* (Sivaraman et al., 2020). Compared to a relatively high prevalence of 81.88% from the previous study, only 48.88% of the *E. coli* isolates from the present study was ESBL- producers. Although we could identify only one non-ESBL isolate of *K. pneumoniae* and the rest being *K. oxytoca* and *K. aerogenes* in the present study, ESBL-producing *K. pneumoniae* accounted for 18.18% with no traces of other *Klebsiella* species in the study conducted previously (Sivaraman et al., 2020). Interestingly, both the studies were conducted in the same sites and involved the representation of same fish varieties. We speculate season and temperature has a tremendous role in determining the bacterial diversity in aquatic environment thereby influencing the bacterial diversity in fish gut and skin. High prevalence of opportunistic and pathogenic bacteria like *E. coli* and *Klebsiella* were reported from the coastal waters of Andaman in monsoon season (August) when compared to non-monsoon (January) (Gobalakrishnan et al., 2014). Also, an aquaculture-based study from East Kolkata, India had shown a higher prevalence of these pathogens in monsoon and post monsoon seasons and this was attributed to the mixture of urban sewage and rainwater runoff from adjacent areas during this season (De et al., 2020). In terms of phylogenetic classification, the majority of *E. coli* isolates from this study belonged to group A irrespective of the carriage of $bla_{\text{CTX-M}}$ genes. Equal representation of phylogroups B1 and D have also been noted in the present study. We had observed similar phylogenetic grouping with two dominant phylogroups B1 and A in a study conducted earlier (Sivaraman et al., 2020). *E. coli* is grouped into three major categories: the commensal strains, known to be an integral part of the intestinal microflora, the intestinal pathogenic strains and extra-intestinal pathogenic strains (Pitout, 2012). It has been shown previously that the majority of commensal strains belong to the phylogenetic group A and B1 (Katouli, 2010) while extra-intestinal

Table 3
Phylogroups, resistance genes and plasmid replicon types of ESBL-producing *E. coli* strains from fish.

Isolate ID	Phylogenetic group	Resistance genes	Plasmid replicon type
WP5–1	A	*CTX-M-15, qnrS, tetA, sul1, sul2*	Y
WP5–2	E	*CTX-M-15, qnrS, oqxB, aac(6)-Ib-cr,*	Y
WP5–3	C	*CTX-M-15, OXA-1-like, qnrS, tetA, aphA1-1AB,*	FIB, P
WP5–4	A	*CTX-M-15, TEM, qnrB, qnrS, oqxB, tetA,*	FIB
WP5–5	A	*CTXM-15, qnrB, qnrS, oqxB, sul1*	FIA
WP5–6	A	*CTX-M-15, qnrS, oqxB, aac(6)-Ib-cr, sul1*	FIA
WP5–7	B1	*CTX-M-15, TEM, qnrB, qnrS, oqxB, tetA,*	FIB
WP5–8	A	*CTXM-15, qnrB, qnrS, tetA, sul1, sul2*	FIA
WP5–9	A	*CTX-M-15, qnrB, qnrS, oqxB,*	FIB
WP5–10	C	*CTX-M-15, TEM, qnrB, qnrS, oqxB, tetA, sul1, sul2*	FIB, X
WP5–11	D	*CTX-M-15, DHA, qnrB, qnrS, oqxB, tetA, sul1, sul2*	FIA, FIB, P, A/C
WP5–12	D	*CTX-M-15, qnrB, qnrS, oqxB*	FIA, FIB, X, A/C
WP5–14	A	*CTX-M-gp-1, qnrB, qnrS, oqxB, tetA, sul1, sul2*	nt[a]
WP5–15	A	*CTX-M-15, qnrB, qnrS, oqxB, aac(6)-Ib-cr, tetA*	nt
WP5–16	A	*CTX-M-15, qnrB, qnrS, oqxB, sul1, sul2*	FIA, FIB, A/C, I1-Iγ
WP5–17	A	*CTX-M-15, qnrS, oqxB, aac(6)-Ib-cr, tetA*	nt
WP5–18	B1	*CTX-M-15, qnrS, oqxB, aac(6)-Ib-cr, tetA*	nt
WP5–19	E	*CTX-M-15, qnrB, qnrS, oqxB*	FIA, FIB, P, A/C
WP5–20	F	*CTX-M-15, qnrS, oqxB, aac(6)-Ib-cr,*	FIA, FIB
WP5–21	F	*CTX-M-15, qnrS, aac(6)-Ib-cr*	FIA, FIB
WP5–22	D	*CTX-M-15, qnrS, oqxB*	FIA, I1-Iγ
WP5–23	B1	*CTX-M-15, SHV, qnrB, qnrS, oqxB, aac(6)-Ib-cr,*	I1-Iγ

a nt- no replicon detected.

Phylogroups A, B1 and E typically represent commensal *E. coli* strains (intra-intestinal) while D consist of extra-intestinal strains capable of causing infections when virulent.

Author Statement.

Sudha Sajeev: Original draft writing, investigation, formal analysis, data curation and editing.

Muneeb Hamza: Investigation, formal analysis and review.

Vineeth Rajan: Investigation, formal analysis and review.

Ardhra Vijayan: Investigation and review.

Gopalan Krishnan Sivaraman: Conceptualization, and fund acquisition.

Bibek R Shome: Conceptualization, fund acquisition.

Mark A Holmes: Conceptualization, fund acquisition.

pathogenic strains belong to B2 and D (Dale and Woodford, 2015). There are several reports stating that the extra-intestinal tract pathogens causing urinary tract infections and diarrhea may also belong to phylogenetic group A (Khairy et al., 2019; Piatti et al., 2008; Adib et al., 2014). The association between phylogroups and virulence factors in determining pathogenicity of a strain is reported only in a few studies (Lee et al., 2010; Picard et al., 1999). Concerning the ESBL genotypes, CTX-M- group 1, 2, 8, 9 and 25 enzymes (mainly CTX-M-15, CTX-M-16, CTX-M-27) are known for their capability to hydrolyze cefotaxime and also ceftazidime (Bonnet, 2004; Nicolas-Chanoine et al., 2008). In the present study, we observed dominance of CTX-M-15 and a lower prevalence of other beta-lactamase enzymes such as TEM and SHV. Studies have reported CTX-M as the most predominant beta-lactamase mechanism found in ESBL-*E. coli* in both animals and humans (Livermore et al., 2007). The predominance of CTX-M-15 in ESBL isolates indicated a selective advantage as it could hydrolyze cefotaxime, ceftazidime and aztreonam. Dominance of CTX-M-15 has been reported by Ramadan et al., 2019, and they have highlighted the importance of these two gene variants (CTX-M-1 and CTX-M-15) in terms of phenotypic resistance to cefotaxime, ceftazidime and aztreonam. The worldwide dominance of CTX-M-15 and its association with the pandemic O25 and ST131 *E. coli* clone has already been documented (Lau et al., 2008). The clonal dissemination of *E. coli* strains carrying $bla_{\text{CTX-M-15}}$ in various habitats is a matter of concern. Besides CTX-M genes, tetracycline resistance gene *tetA* was detected in 33% of *E. coli* and 4% of *Klebsiella* sp. Co-existence of ESBL genes and other antibiotic resistance genes along with plasmid-mediated *tetA,* responsible for efflux-pump mechanism may enhance the persistence of plasmid-bearing strains in an environment contaminated with antimicrobial compounds (Çiçek et al., 2022). The isolates that had shown phenotypic resistance or intermediate MIC values for carbapenems did not show the presence of any carbapenem resistance genes tested. There are a few studies that have shown drawbacks in accurate prediction of CRE in automated antimicrobial susceptibility testing systems calling for more appropriate molecular methods (Pollett et al., 2014; Solanki et al., 2014; Baran and Aksu, 2016). Emergence of disinfectant resistance in bacteria can be attributed to the expression of disinfectant resistance genes through the mediation of efflux mechanism. Horizontal gene transfer through conjugation is one of the main reasons for the emergence of new resistant pathogens and their dissemination (Partridge et al., 2018). The possibility of co-transfer of beta-lactamase genes and QAC genes through horizontal gene transfer had been emphasized previously (Li et al., 2017). In the present study, incidence of *E. coli* carrying QAC genes is much higher than that in *Klebsiella* strains. All the chromosome-encoded QAC resistance genes tested were detected in all *E. coli* isolates irrespective of their antimicrobial resistance status. *qacEΔ1* is the most frequent plasmid-mediated QAC resistance gene in *E. coli*, with 59% of ESBL isolates and 26% of non-ESBL *E. coli* carrying it. According to the studies conducted by Wu et al. (2015), the majority of Gram-negative bacteria harboured *qacEΔ1* in class I integrons. Earlier studies have concurred that there is no correlation between the presence of *qacE/qacΔ1* genes and resistance or increased MIC values to disinfectants. Increased MIC of a strain to disinfectants particularly benzalkonium chloride, cetyl-trimethylammonium bromide or ethidium bromide cannot be attributed to presence or absence of *qacE/qacEΔ1* nor the multi drug resistance status of the isolate (Kücken et al., 2000). QAC resistance genes such as *qacE, qacF* and *qacG* were completely absent in the *E. coli* and *Klebsiella* isolates studied. Lower incidence of these disinfectant resistance genes has been reported earlier; one among the 37 clinical isolates of *P. aeruginosa* studied, carried *qacE* (Kücken et al., 2000) while *qacF* and *qacG* was detected in 1.8% and 0.4% of *E. coli* isolated from retail meats, respectively (Zou et al., 2014). The use of disinfectants is unavoidable in various stages of food manufacturing and processing industry. The widespread use of disinfectants/quaternary ammonium compounds in different settings such as health care, food processing, agricultural and other industries may contribute to the emergence of antimicrobial resistant bacteria. Disinfectant resistant bacterial strains with reduced effectiveness would promote food-borne diseases, challenging our public health care system (Tong et al., 2021).

IncF-type-plasmids were the dominant group detected among $bla_{\text{CTX-M-15}}$ carrying *E. coli* strains from this study. Overall, 50% of the ESBL-producing isolates harboured IncFIB-type and 45% of the non-ESBL *E. coli* were also positive for IncFIB-type. A few of the beta-lactamase producing isolates carrying IncF-type-plasmids also showed replicons of A/C, X, P and I1-Iγ. Plasmids are the known carriers of ESBL genes (Li et al., 2019) and co-existence of CTXM-1/CTXM-15 genes on these plasmids are highly plausible given the fact of transmissibility through horizontal gene transfer. Presence of multi-replicon plasmids in $bla_{\text{CTX-M-15}}$ carrying *E. coli* isolates had previously been reported (Zurfluh et al., 2015). F replicons are narrow host-range plasmids that have adapted well in *E. coli* and carry out conjugation easily (Marcadé et al., 2009). IncF plasmids, with its heterogeneous nature and being diffusion-

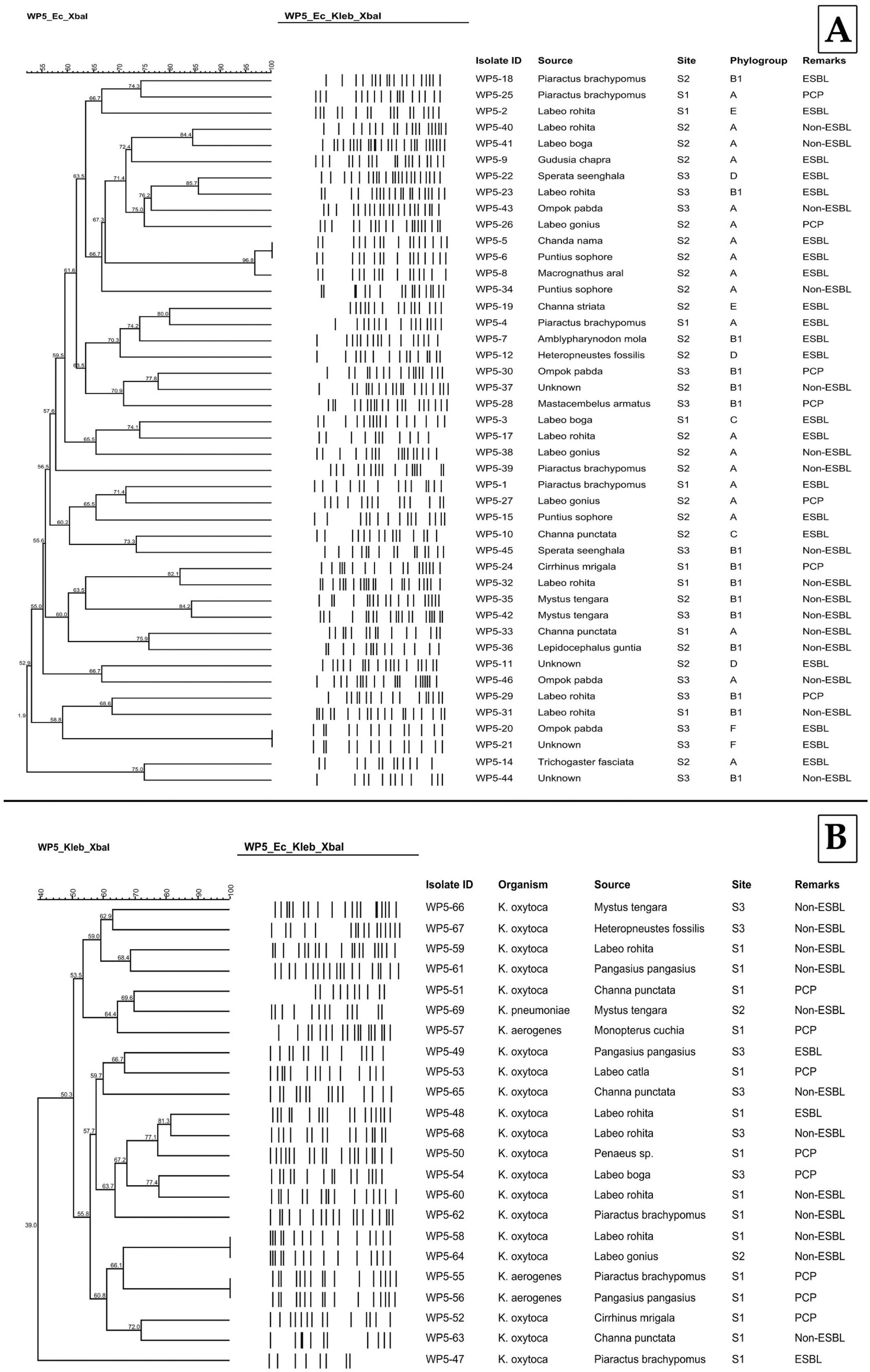

Fig. 4. Dendrograms showing cluster analyses of *Xba1*-PFGE patterns of *E. coli* (A) and *Klebsiella* (B).

specific, could also acquire $bla_{CTX-M-15}$gene in *E. coli* (Coque et al., 2008). In commensal *E. coli*, IncF families of plasmids are stably maintained and are almost exclusively responsible for the global dissemination of $bla_{CTX-M-15}$ genes. Generally ESBL-producing *E. coli* strains that carry the dominant CTX-M-15 gene belong to highly virulent B2 phylogroup with multi-drug resistant IncFII plasmid (Bevan et al., 2017), while all the *E. coli* isolates with $bla_{CTX-M-15}$ from the present study belonged mainly to phylogroup A and only 50% of the strain showed multi-drug resistant IncFI-type plasmid. Evidence of $bla_{CTX-M-15}$ gene located on specific plasmids of IncF-type has also been reported by Woodford et al., 2009. In the present study, *E. coli* isolates carrying $bla_{CTX-M-15}$ with IncF type plasmid have shown resistance to quinolones, aminoglycosides and sulfonamides. Extensive recombination events in IncF plasmids has contributed to the presence of multi-replicons and diverse plasmid size, transferability, antimicrobial drug–resistance genes allowing them to evolve and persist in various *E. coli* populations (Levin, 1995; Osborn et al., 2000). In the present investigation, 18% of the beta-lactamase-producing *E. coli* isolates from fish belonged to IncA/C plasmid type. IncA/C group of plasmids has also been reported from other food sources like beef, pork, chicken and turkey, establishes the broad dissemination of this plasmid (Mulvey et al., 2009). Certain plasmid families like IncFII, IncA/C, IncL/M, IncN and IncI1 plasmids are frequently detected among Enterobacteriaceae from different origins and sources. These plasmids are known to carry and disseminate specific ESBL and pAmpC genes and are considered as "epidemic resistance plasmids" (Carattoli, 2011). Determination of such plasmid types in Enterobacteriaceae is crucial for understanding epidemiology and designing intervention strategies to manage dissemination. Four of the ESBL-producing *E. coli* and all of the *Klebsiella* isolates could not be assigned to any plasmid types. This may be due to the fact that 1) only limited plasmid replicons can be screened using the Carattoli method; or 2) the isolates are harboring plasmids of an unknown subtype. The PFGE banding pattern obtained for the *E. coli* strains were highly diverse. Forty one unique patterns were derived with <85% similarity suggesting a multi-sectoral influence. The diverse *E. coli* population in an aquatic environment indicates the impact of industrial effluent/ sewage/ hospital waste etc. Similarly, with 19 unique patterns among the 23 isolates, *Klebsiella sp*, also exhibited considerable genetic diversity. The presence of $bla_{CTX-M,}$ antibiotic resistance genes and phylogroup, did not affect the phylogenetic clustering of the isolates. The zoonotic potential of these clonally different ESBL *E. coli* and *Klebsiella* cannot be undermined.

5. Conclusion

In summary, the present study demonstrated that *E. coli* and *Klebsiella* isolated from retail fish samples harboured an array of antibiotic and disinfectant resistance genes. The predominant ESBL genotype in these fish-sourced isolates was CTX-M-15, as is the case in isolates from clinical and various other settings from this geographical area. Our study reiterates that fishes can act as a reservoir of pathogens carrying different beta-lactamase and other multi-drug resistance genes, thus creating an environment for infection and dissemination through clonal expansion. Circulation of such isolates would naturally propagate resistance to other species in an aquatic habitat subsequently spreading to human population. Poor hygienic facilities at landing centres or during handling, storage and transportation would also aid in the spread of infectious multi-drug resistant pathogens. The prevalence of disinfectant resistance and antimicrobial resistance genes in different phylogroups of *E. coli* and *Klebsiella* suggests the adaptation and genome plasticity of these species. With large consumption of antibiotics, it appears likely that fish farms will act as AMR hotspots facilitating the emergence and dissemination of new resistance determinants. Therefore, continuous monitoring and concerted actions are required to mitigate AMR in fisheries and aquaculture settings.

Credit authorship contribution statement

Original draft writing: Sudha Sajeev; Micobiological investigation, formal analysis, data curation, editing and review of the manuscript were performed by Sudha Sajeev, Muneeb Hamza, Vineeth Rajan, Ardhra Vijayan and Gopalan Krishnan Sivaraman; Conceptualization, Fund acquisition, Resources, Supervision of the project: Mark A. Holmes, Bibek R. Shome, and Gopalan Krishnan Sivaraman.

Declaration of submission

All the authors approved the submission of this manuscript. All the authors declare that this manuscript has not been submitted anywhere and is not under consideration by any journals. If the manuscript is published, it will not be submitted anywhere in the same form, in English or in any other language.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The necessary data have been attached as the supplementary file. In case of any further clarifications/data are required, please feel free to contact the corresponding author.

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Appendix A. Supplementary data

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